

Year 10 Science
Semester 1 Examination 2017 - Revision
Astronomy + Physical Science

Astronomy

1. Complete the following table.

TERM	DEFINITION
light year	Distance light travels in one year.
constellation	Group of stars forming a pattern.
galaxy	Collection of stars held together by gravity.
nebula	Gas cloud.
supernova	Massive explosion of a massive star near the end of its life.
red giant	Medium stars swell in size as they run out of fuel.
black hole	Very dense and huge gravity. Light can't escape.
sunspot	Dark areas on the surface of the Sun - sites of large eruptions.
corona	Upper atmosphere of the Sun.
fusion	Process producing energy in stars - hydrogen becomes helium.

2. Light travels at 300,000 km/sec. A star is 97.6 light years away. We can only travel at about 15 - 20 km/second with our best rockets.

(a) With our current technology, are we likely to be able to travel to this star in the near future? Give a few reasons for your answer.

YES ☒ NO (Circle your answer.)

Reasons:

- Spaceship is far too slow.
- The distance is huge and will take hundreds of years to reach.
- Will die well before we get there.

(b) Suggest **one way** we might be able to overcome the time factor of travelling to other galaxies, given it could be hundreds of years.

- Develop craft that can travel near the speed of light.
- Freeze people and wake them up hundreds of years later.
- Learn how to warp space and fold it on itself - makes the distance much shorter.

3. (a) What is the major element found in the Sun? hydrogen

(b) What happens to this element so that the Sun produces its energy?

Hydrogen atoms collide and become helium - needs 1 million °C to do this. Energy is released.

(c) The Sun loses 5 million tonnes of mass per second. According to Einstein's equation ($E = mc^2$), what does this mass become?

The lost mass becomes the energy that the Sun emits.

4. How is the surface temperature of a star related to its colour?

TEMPERATURE	COLOUR
very hot	blue
cool	red
moderate	yellow

5. (a) About how old is our Sun (in millions of years)? 5000 million years

(b) How much longer will it "live"? 5000 million years

6. (a) Draw a flow diagram (with arrows) to show the stages of our Sun's remaining life as it runs out of fuel.

Sun → red giant → nova → white dwarf → black dwarf

(b) Do similar flow diagrams for the following types of stars.

Huge star → red supergiant → supernova → neutron star

Giant star → red supergiant → supernova → black hole

(c) The difference between a **nova** and a **supernova** is the size of the explosion that takes place. What happens to a star to cause it to explode?

Star starts to run out of fuel and (ultimately) collapses on itself.

7. (a) Describe the size of the gravity near a black hole.

huge

(b) Why does a black hole appear black?

Gravity is so big that light can't escape from it.

8. Complete the following matrix to compare the Big Bang and Steady State Theories, using ticks to show which statements apply to one, both or neither theory.

Statement	Big bang theory	Steady state theory
The universe has no beginning.		Y
The universe began with a single point called singularity.	Y	
The universe is expanding.	Y	
The universe always looks the same.		Y
The red shift in the spectrum of visible light coming from stars and galaxies provides evidence for the theory.	Y	
New stars and galaxies are created to replace those that move away due to expansion of the universe.		Y
This theory explains the amount of helium in the universe.	Y	
This theory is supported by the measurement of the current temperature of the universe (about -270°C).	Y	
This theory was first supported by a Catholic priest.	Y	
The theory will never be proven incorrect.		
An end was put to this theory in 1965.		Y

9. Why are the constellations we see now so different from the way they were many centuries ago?

Stars have exploded at various times, so some have disappeared.

10. How have scientists gained their knowledge of the life and death of stars if the processes involved take millions of years to occur?

Light from distant stars takes millions of years to get to us, so we see them as they were at that time. Closer stars appear different, so we can see them at different stages.

11. What is the difference between a neutron star and a black hole?

Neutron star - remains of a massive star; dense; composed of neutrons.

Black hole - remains of a giant star; very dense; huge gravity; light can't escape.

12. What makes stars group together to form galaxies?

Gravity pulls all of the matter and stars together.

13. The Doppler effect is most commonly associated with the changing pitch of a sound as its source moves past you. For example, the pitch of the noise made by a speeding train increases as it approaches you and decreases as it moves away from you. Explain how the Doppler effect is relevant to the study of the universe.

The spectra of light from stars contain dark lines caused by the absorption of certain colours by substances in the star.

As the stars move away our solar system at high speed, the Doppler effect causes the lines to move towards the red end of the spectrum.

As the stars move towards our solar system at high speed, the Doppler effect causes the lines to move towards the blue end of the spectrum.

14. What is cosmic microwave background radiation and why does it exist?

Cosmic microwave background radiation is radiation that was released 380 000 years after the Big Bang.

Physical Sciences

1. A man drives his car 800 m due East and then turns North, driving 300 m before stopping at a petrol station. It took him 120 s to get there.

(a) What distance he has driven.

$$d = 800 + 300 = 1100 \text{ m}$$

(b) Calculate his average speed for the journey.

$$sp = ?$$

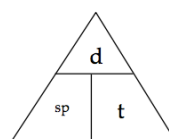
$$d = 1100 \text{ m}$$

$$t = 120 \text{ s}$$

$$sp = \frac{d}{t}$$

$$= \frac{1100}{120}$$

$$= \underline{9.17 \text{ m/s}}$$



2. A balloon filled with hydrogen gas has a velocity of 40 m/s East. How long will it take to go 450 m?

$$sp = 40 \text{ m/s}$$

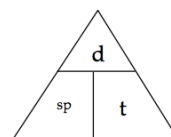
$$d = 450 \text{ m}$$

$$t = ?$$

$$t = \frac{d}{sp}$$

$$= \frac{450}{40}$$

$$= \underline{11.2 \text{ s}}$$



3. What is the average speed of a plane if it flies 2700 km in 2.3 hours? Give the answer in km/h.

$$sp = ?$$

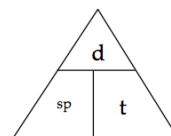
$$d = 2700 \text{ km}$$

$$t = 2.3 \text{ hr}$$

$$sp = \frac{d}{t}$$

$$= \frac{2700}{2.3}$$

$$= \underline{1174 \text{ km/h}}$$



4. A top sprinter reaches 11.2 m/s within 3.15 s of the start of a race. Calculate the average acceleration.

$$v = 11.2 \text{ m/s}$$

$$u = 0 \text{ m/s}$$

$$a = ?$$

$$t = 3.15 \text{ s}$$

$$a = \frac{v - u}{t}$$

$$= \frac{11.2 - 0}{3.15}$$

$$= \underline{3.56 \text{ m/s}^2 \text{ forwards}}$$

5. Calculate the final velocity of a car initially travelling at 10 m/s that accelerates at 1.4 m/s^2 for 10 s.

$$v = ?$$

$$u = 10 \text{ m/s}$$

$$a = 1.4 \text{ m/s}^2$$

$$t = 10 \text{ s}$$

$$v = u + at$$

$$= 10 + 1.4 \times 10$$

$$= \underline{24 \text{ m/s forwards}}$$

6. Determine the average acceleration of a car that increases its velocity from 7.0 m/s to 34 m/s in 9.0 s.

$$v = 34 \text{ m/s}$$

$$u = 7 \text{ m/s}$$

$$a = ?$$

$$t = 9.0 \text{ s}$$

$$a = \frac{v - u}{t}$$

$$= \frac{34 - 7}{9.0}$$

$$= \underline{3.0 \text{ m/s}^2 \text{ forwards}}$$

7. A truck moving at 36 m/s is decelerated uniformly at 1.5 m/s². What is its final velocity after 5.0 s?

$$V = ?$$

$$u = 36 \text{ m/s}$$

$$a = -1.5 \text{ m/s}^2$$

$$t = 5.0 \text{ s}$$

$$V = u + at$$

$$= 36 + (-1.5) \times (5.0)$$

$$= \underline{28.5 \text{ m/s forwards}}$$

8. A car of mass 1530 kg is travelling at 70.0 km/h (19.4 m/s) along Hepburn Avenue when the driver has to brake sharply to avoid a small child who suddenly appeared on a bike crossing the road.

- (a) If it took 3.5 s to stop the car, calculate the deceleration of the car.

$$V = 0 \text{ m/s}$$

$$u = 19.4 \text{ m/s}$$

$$a = ?$$

$$t = 3.5 \text{ s}$$

$$a = \frac{V - u}{t}$$

$$= \frac{0 - 19.4}{3.5}$$

$$= -5.54 \text{ m/s}^2$$

$$\therefore a = \underline{5.54 \text{ m/s}^2 \text{ backwards}}$$

- (b) Calculate the average force exerted by the brakes.

$$F = ?$$

$$m = 1530 \text{ kg}$$

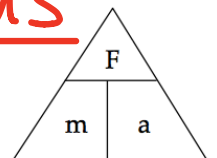
$$a = -5.54 \text{ m/s}^2$$

$$F = ma$$

$$= (1530) \times (-5.54)$$

$$= -8476 \text{ N}$$

$$\therefore \underline{F = 8476 \text{ N backwards}}$$



- (c) The driver of the car was an engineer and had a hard-hat sitting on the back shelf. Describe what happened to the hard-hat in this situation and explain why it took place.

- Hard-hat has inertia.
- It continues to move forwards while the car stops.

9. (a) Explain the difference between **mass** and **weight**. Give an example to help your explanation.

Mass - a measure of the matter in an object.

Weight - force due to gravity on an object.

- (b) The acceleration due to gravity on Jupiter is 22.5 m/s^2 . Calculate the weight of a 450 kg object on Jupiter.

$$F_w = ?$$

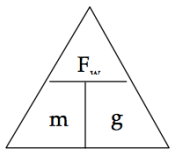
$$m = 450 \text{ kg}$$

$$g = 22.5 \text{ m/s}^2$$

$$F_w = mg$$

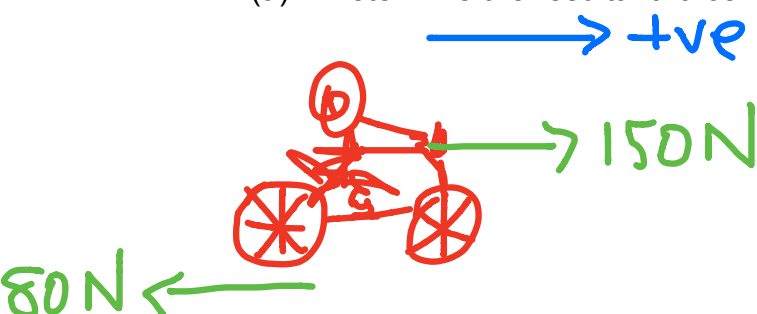
$$= 450 \times 22.5$$

$$= \underline{10125 \text{ N}}$$



10. A 70.0 kg cyclist is applying a force of 150 N forwards but is working against the wind, producing a friction force of 80 N opposing the movement.

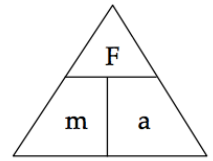
- (a) Determine the resultant force for this situation.



$$\begin{aligned}\Sigma F &= 150 - 80 \\ &= \underline{70 \text{ N forwards}}\end{aligned}$$

(b) Calculate the acceleration of the cyclist.

$$\begin{aligned}\Sigma F &= 70 \text{ N} & a &= \frac{\Sigma F}{m} \\ m &= 70 \text{ kg} & &= \frac{70}{70} \\ a &= ? & &= 1.0 \text{ m/s}^2 \text{ forwards}\end{aligned}$$

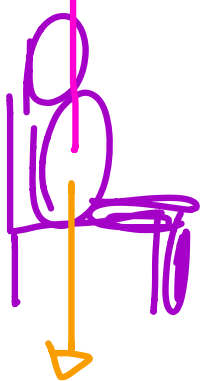


(c) It is possible for the rider to move at constant velocity. In terms of the forces acting, explain how this is possible.

- Yes
- Force forwards = force backwards due to friction and the wind.

Reaction

11. You are sitting on a seat at the moment. Explain why you don't fall through the chair and are supported by it.

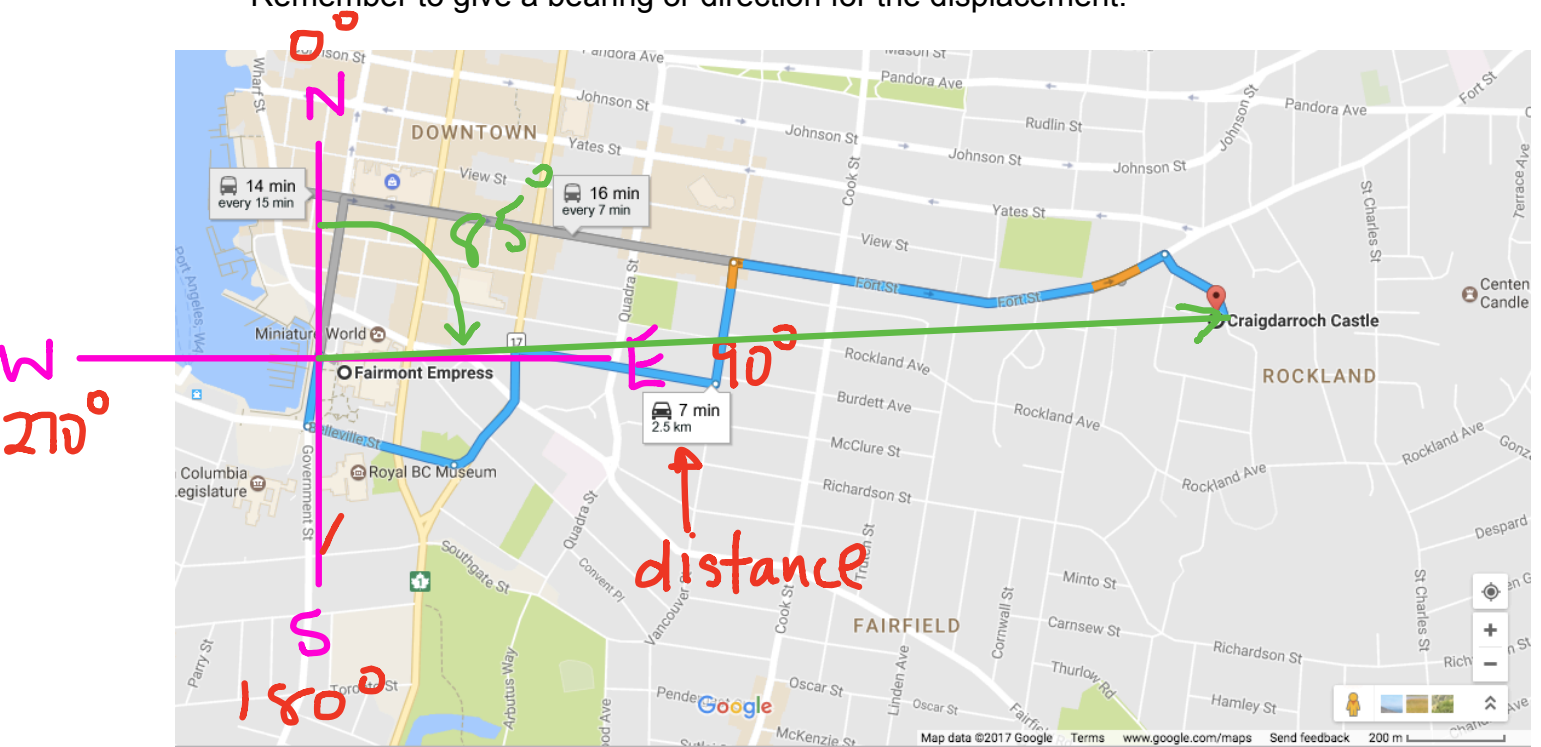


F_{weight}

- Gravity pulls you downwards - weight force F_w .
- Chair exerts an equal and upward reaction force R .

12. Use the scale in the map below to work out the **distance and displacement** from the Fairmont Empress hotel to Craigdarroch Castle. Use either the blue or grey route.

Remember to give a bearing or direction for the displacement.



distance = 2.5 km

displacement = 1.8 km

bearing = 85°